

A brief and highly incomplete guide to the history and philosophy of statistics

(Aka – why is this all so bldy confusing?!)**

**Hannah Wauchope
EEA Week 9
2026**

‘Statistics’ as a field

- **Is very young!**
- **Nearly everything we now consider as ‘statistics’ has been developed in the last century**

‘Statistics’ as a field

- **Is very young!**
- **Nearly everything we now consider as ‘statistics’ has been developed in the last century**
- **To do with how we collect, analyze, and interpret data *under uncertainty***

If we had all information on everything, statistics wouldn’t exist

Statistical Questions

- **“We have this data, does it represent current reality well?”**
- **“We have this data, can it predict the future well?”**

Statistical Questions

- “We have this data, does it represent current reality well?”
- “We have this data, can it predict the future well?”
- There is no ‘one true way’ of doing statistics. It’s a bunch of people making up methods.

This means we need to *always* be thinking critically. If you can fight the fear of scary words and numbers, keep your head and keep thinking logically, you *will* be good at statistics.

Today – What will I talk about?

- **Why there are weird names for statistical tests**
- **Why some statistical tests are actually just variations on a theme of other statistical tests**
- **Different statistical philosophies (with a brief detour to caution about P-Values)**

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Statistical Approaches

- **Why all the weird names?**
- **It's just people inventing things, we most often name the tests after the people that invented them.**

Let's take ANOVA as an example

- **What should ANOVA actually be called?**

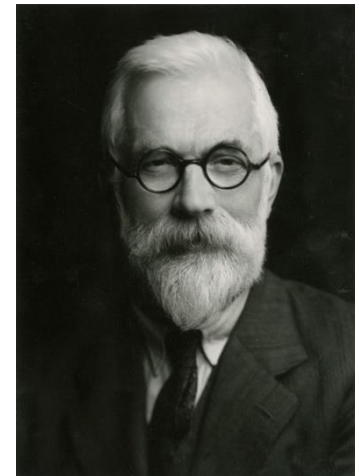
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- **What should ANOVA actually be called?**

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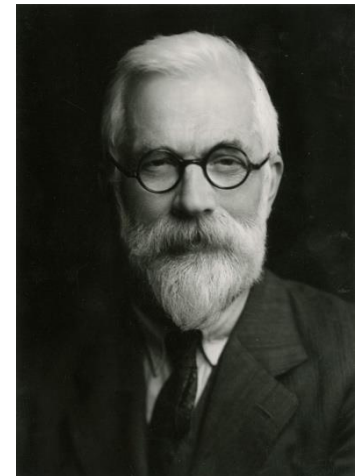
- **It's just a test invented by this guy**



Ronald Fisher (1890-1962)

Ronald Fisher (ANOVA inventor)

- **“A genius who almost single-handedly created the foundations for modern statistical science”**
- **Very recent: people working today worked with or were taught by Ronald Fisher**



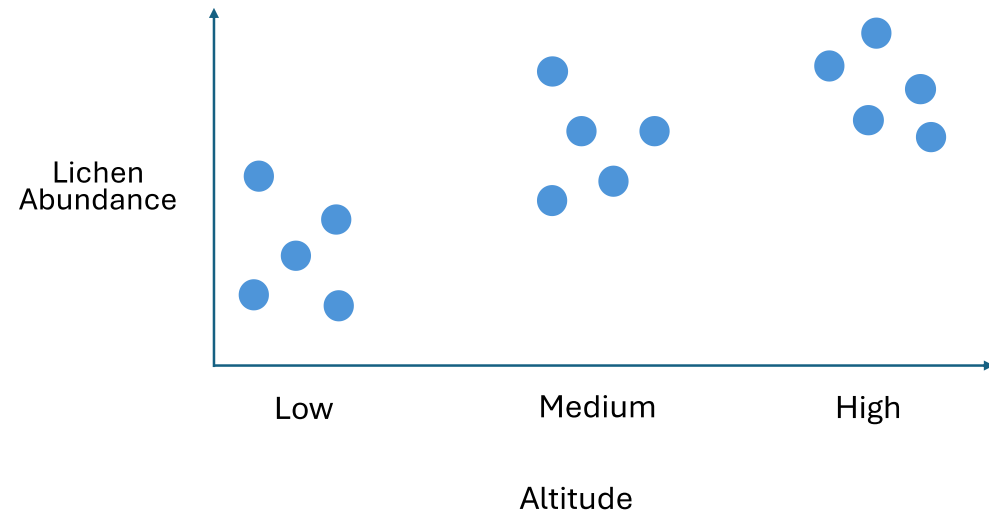
Ronald Fisher (1890-1962)

Why all the Assumptions?

- **Fisher designed ANOVA to work with a particular type of data – ideally a simple dataset collected in an experimental set up.**
- **If your data doesn't sort of meet the assumptions, his method doesn't make sense.**

Why all these extra bits?

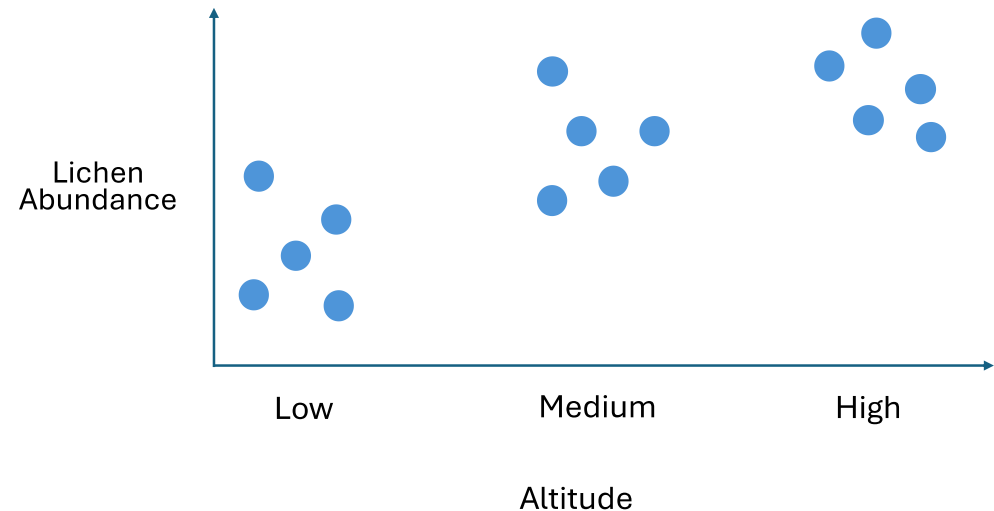
- Fisher designed ANOVA to just say “is there a difference between groups.”
- But what if you have lots of groups? And want to know *which* groups are different?



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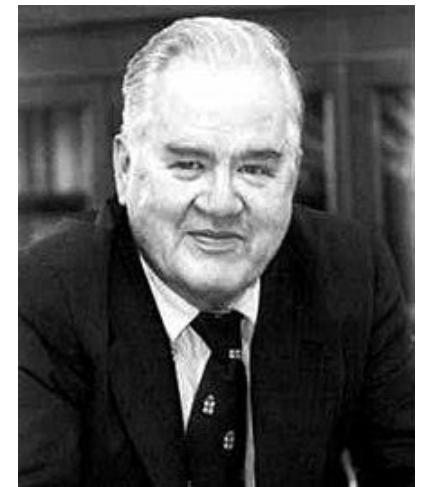
ANOVA might say these groups are significantly different. But looking at our graph, we suspect that not all groups are different from each other



Why all these extra bits?

- Fisher designed ANOVA to just say “is there a difference between groups.”
- But what if you have lots of groups? And want to know *which* groups are different?
- ***Enter Tukey with his Post-Hoc Test***

(Note: ‘Post-Hoc’ is just a fancy way of saying “the original test didn’t do what we wanted, so we’re going to do a follow up test”)



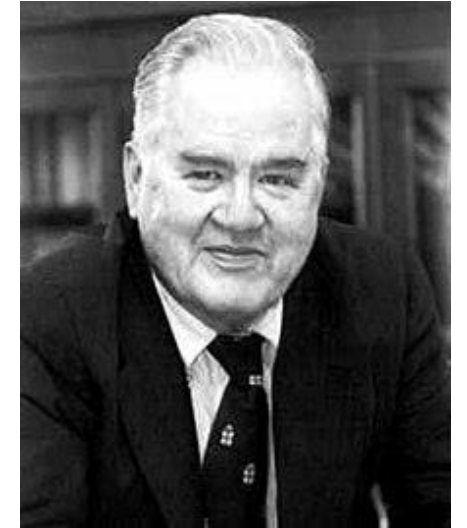
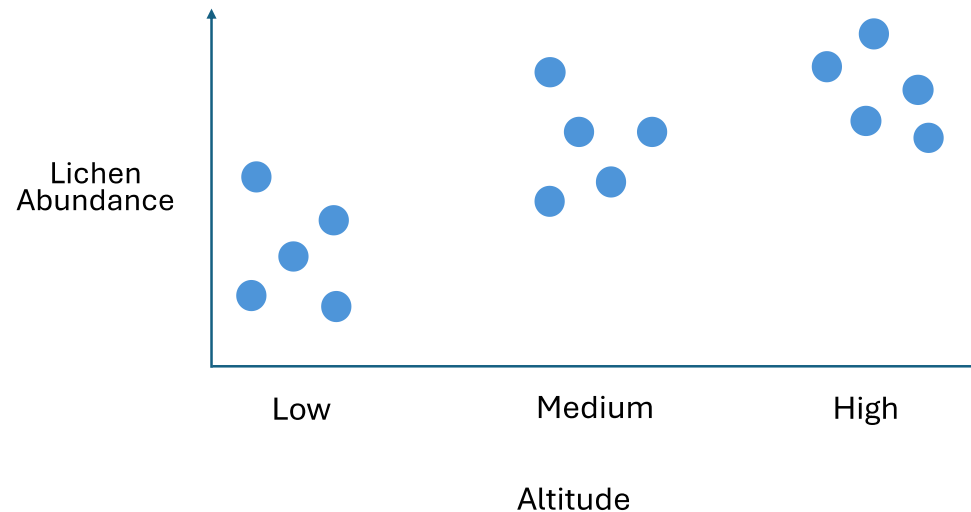
John Tukey (1915-2000)

Why all these extra bits?

- ***Enter Tukey with his Post-Hoc Test***

ANOVA says significant difference between groups.

Tukey test says Low and Medium are different, Low and High are different, but Medium and High are not.



John Tukey
(1915-2000)

I'll stop here on this example

- **Moral of the story: complex names do *not* mean complex tests. Mathematicians and Statisticians like explaining things in complex ways.**

MY CHALLENGE TO YOU: Any time you encounter a new statistical test with a scary name, do some googling and then attempt to rename it to something (overly simplified) and descriptive

E.g. ANOVA → The Test to See if Groups Are Different

E.g. Tukeys → The Test to See How Each Group Differs From The Others

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Statistical Approaches

- Often, people invented similar things and we end up with multiple names for essentially the same thing.
- If someone could tidy up all of stats and start again that'd be *great*. But they can't, so here we are.
- For example...

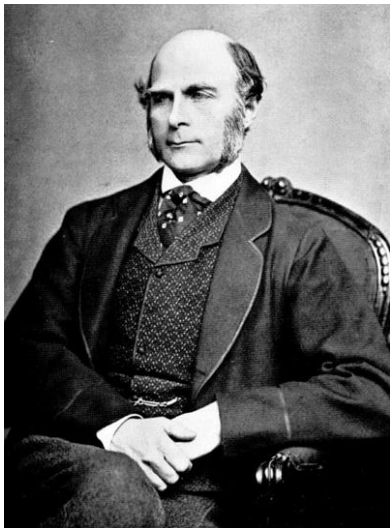
On Linear Models

- **Around the same time (give or take a few decades) that Fisher was inventing ANOVAs, linear regression was being developed by Galton And Pearson**

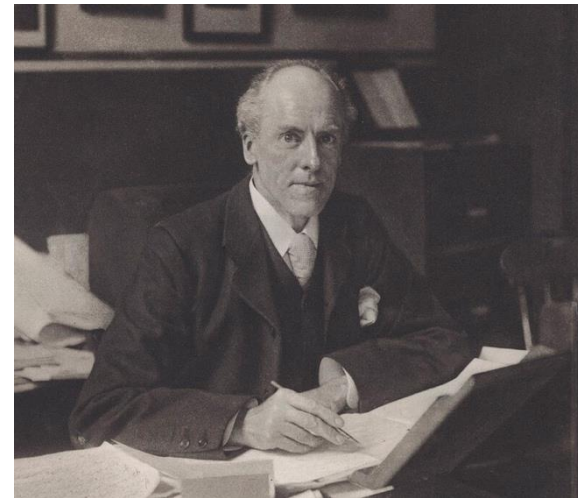
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Hello



Francis Galton (1822-1911)



Karl Pearson (1857-1936)

*Good
Evening*

On Linear Models

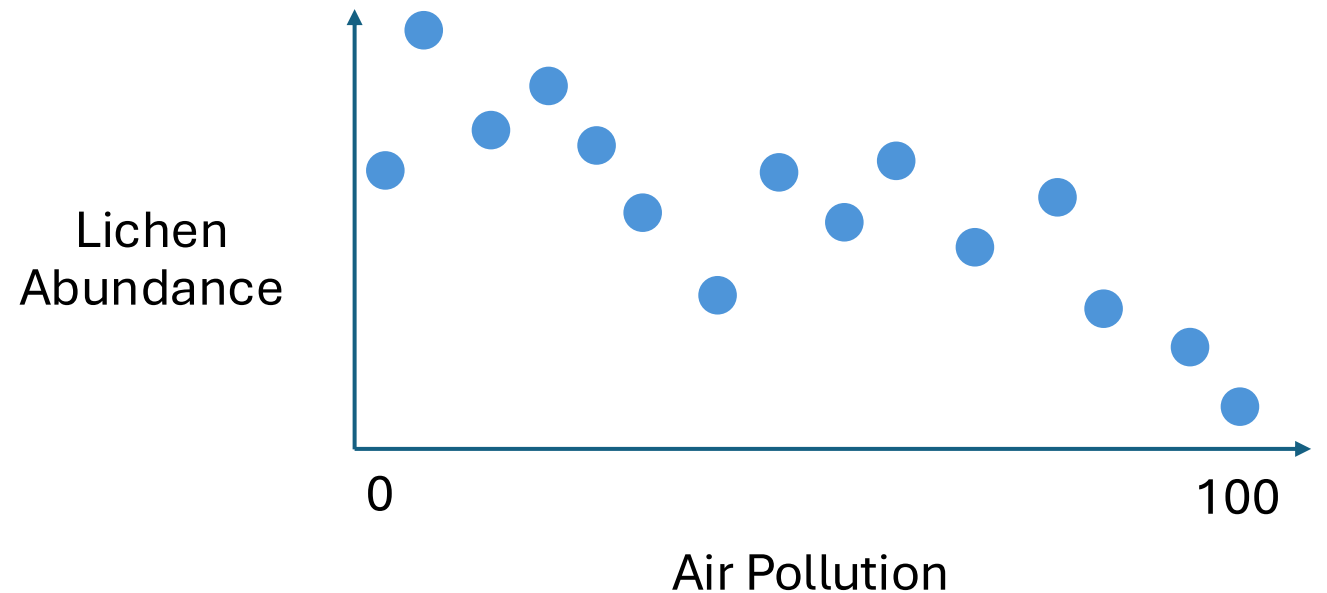
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- **“Can one variable predict another variable?”**

On Linear Models

- **“Can one variable predict another variable?”**
- **Initially for two continuous variables**
- **“Test to see if there is a *simple* relationship between variables (e.g. if one goes up, does the other go down?)**

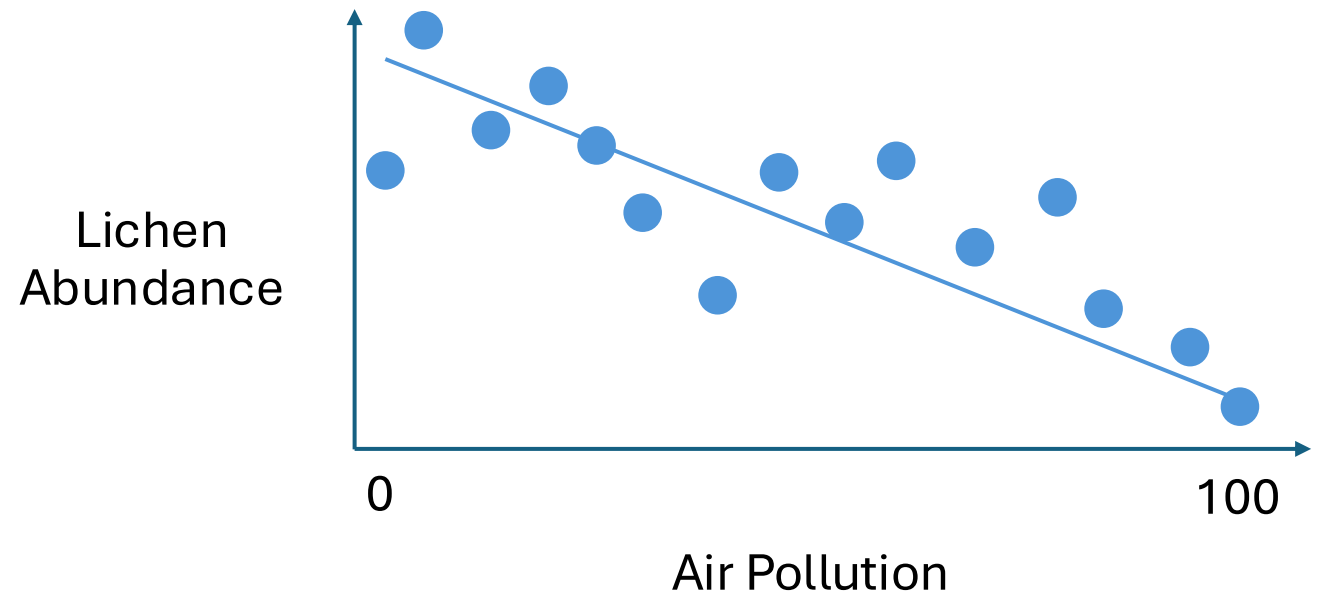
On Linear Models

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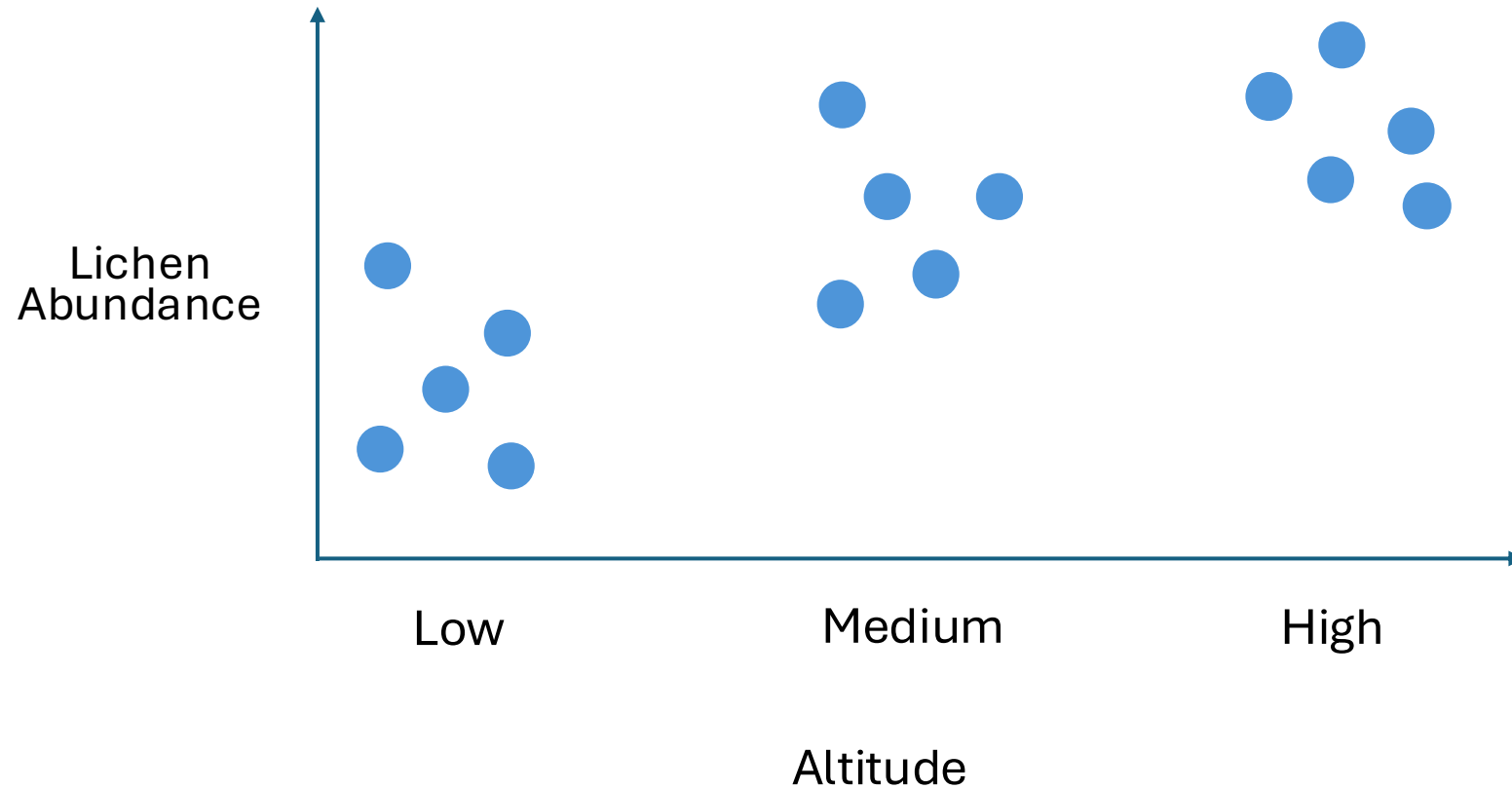
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On Linear Models

- **Since then, linear regression has been expanded and the models we use can also include categorical predictors.**
- **E.g. if we know which group something is in, can we predict its value**
- **This means...**

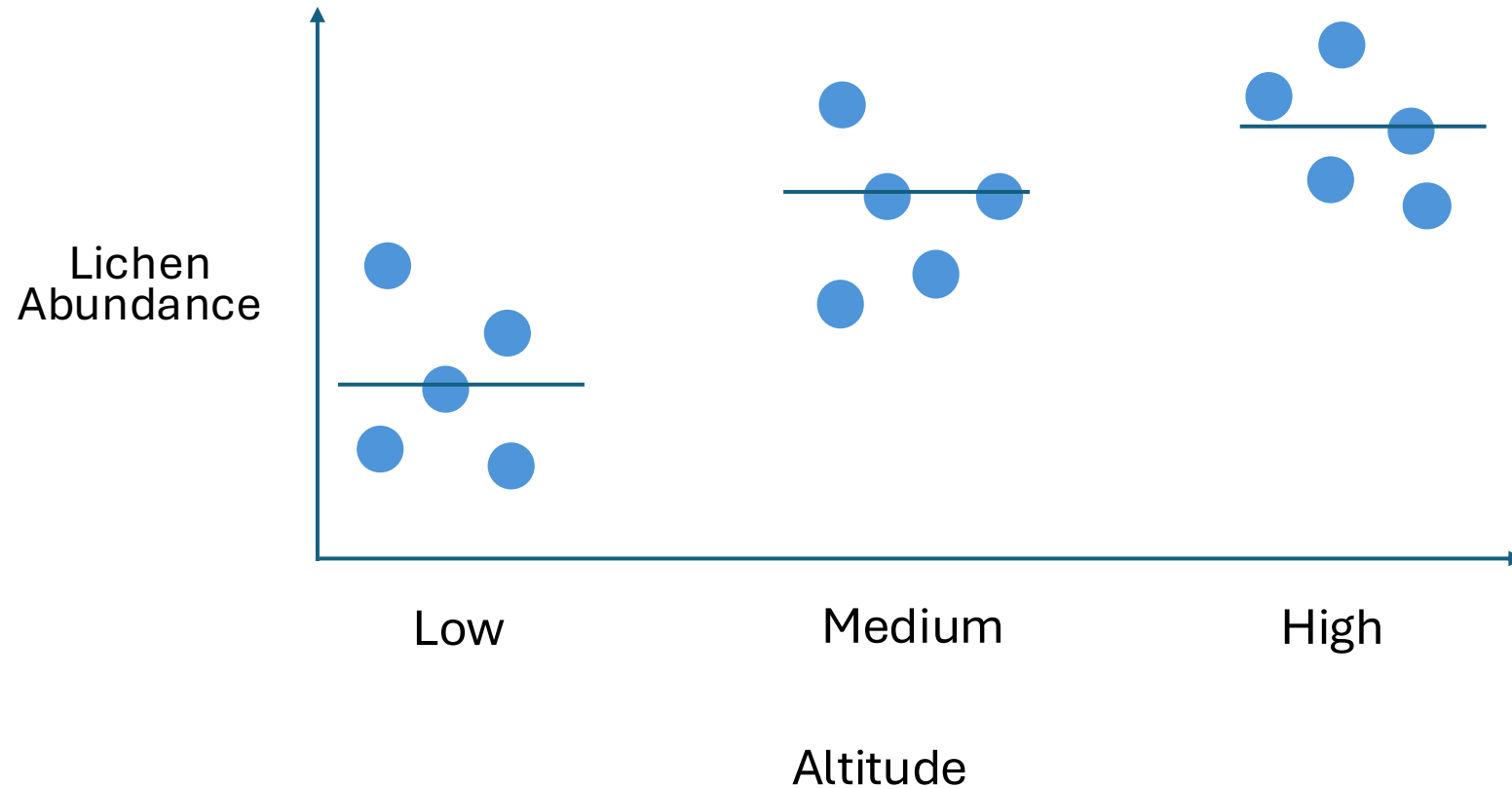
On Linear Models



You could run a linear model:

Abundance $\sim$ Altitude

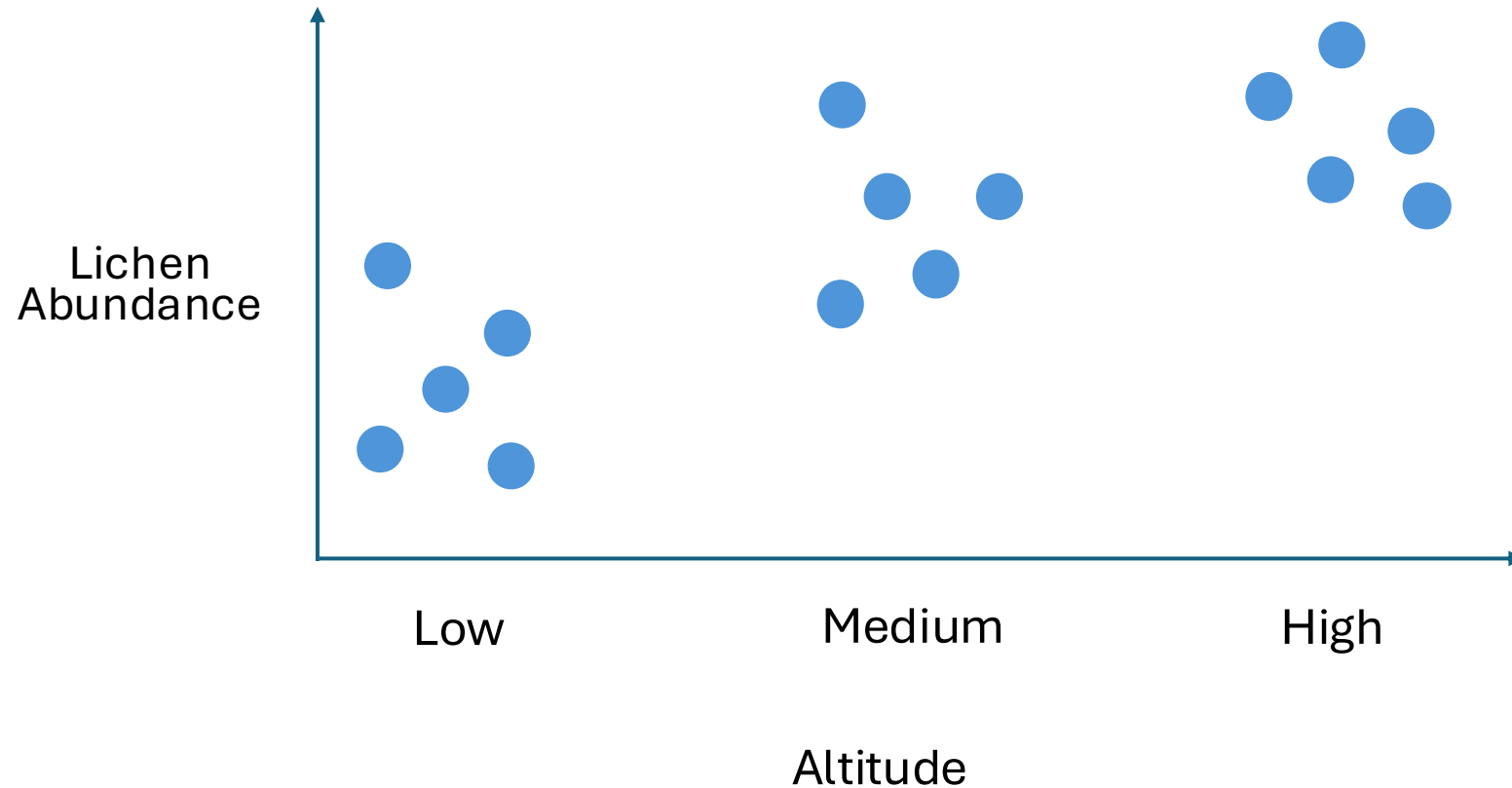
On Linear Models



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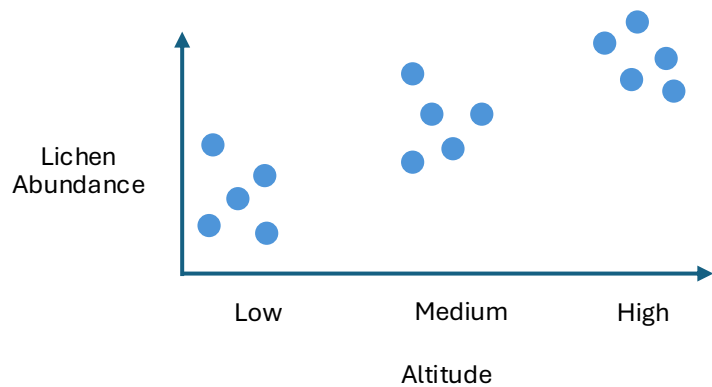
OR

You could run an ANOVA:

$\text{Abundance} \sim \text{Altitude}$

On ANOVA vs Linear Models

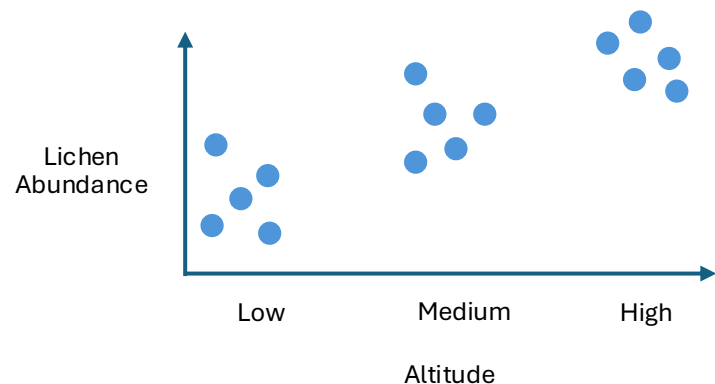
- This means... ANOVA is really just a Linear Model with one categorical predictor
- But it will output slightly information:
 - ANOVA: “Are Groups Different”



E.g. “Yes, lichen abundance is significantly different between groups of different altitudes.”

On ANOVA vs Linear Models

- **ANOVA is really just a Linear Model with one categorical predictor**
- **But it will output slightly information:**
 - **ANOVA: “Are Groups Different”**
 - **LM: “How different is each group to the first group”**



**E.g. “Lichen abundance at Medium Altitudes is 3 higher than Low (significant).
Lichen abundance at High Altitudes is 6 higher than Low (not significant)”**

What about Chi Squared Tests? (χ^2)

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- **ALSO just a test asking if two variables are related**
- **What's the difference? The two variables are categorical, not continuous (more specifically, counts of things within groups)**
 - **E.g. Are certain sea snail species more common on certain types of seaweed**

What about Chi Squared Tests? (x2)

- **ALSO** just a test asking if two variables are related
- **What's the difference?** The two variables are categorical, not continuous (more specifically, counts of things within groups)
 - E.g. Are certain sea snail species more common on certain types of seaweed
- **WHY** is it called something so confusingly different if it's just a variation on a theme?

Blame this guy (Karl Pearson again)



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On other ways of doing Statistics

- **Because the field of statistics is just figuring itself out as it goes, there isn't just one way of doing things...**
- **Final section of this lecture: just opening your eyes to the fact that other ways of thinking exist**
- ***You don't need to understand in depth the following – just to be aware (so if you read something in a paper you're not bamboozled)***

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ANOVA

Linear Model



**Frequentist Approaches, based on null
hypothesis testing**

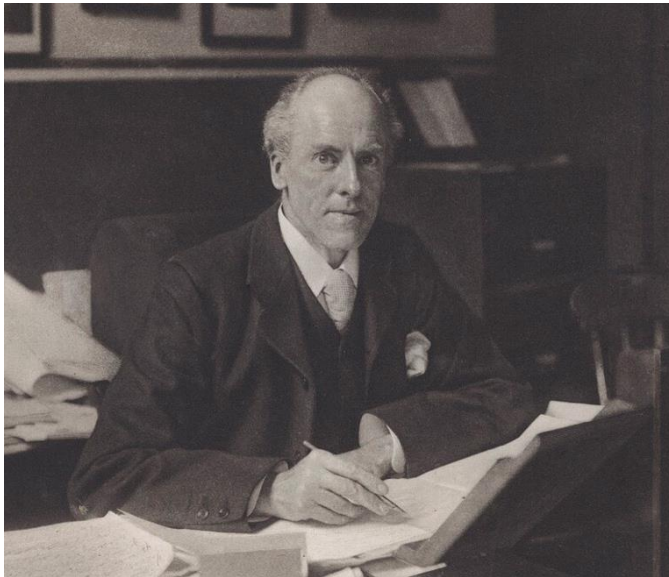
**Will you allow me a moment to make a brief
cautionary warning about P-Values**

P Values

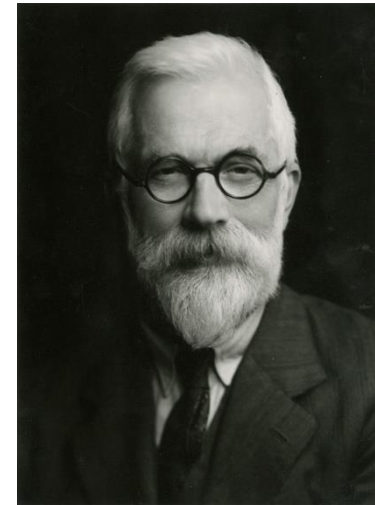
- **“The probability of obtaining the results given the null hypothesis is true”**

P Values

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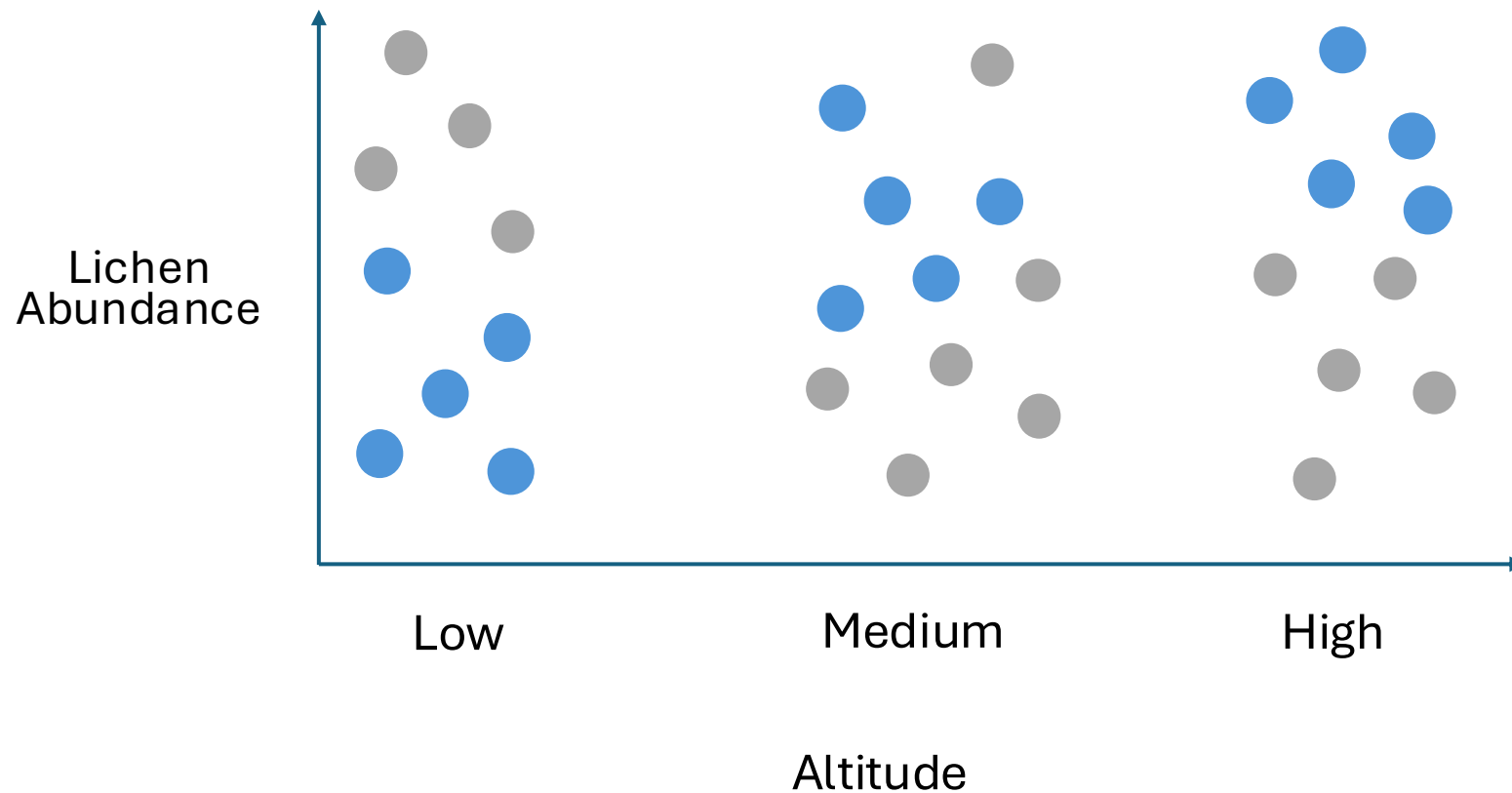
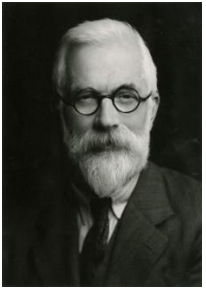
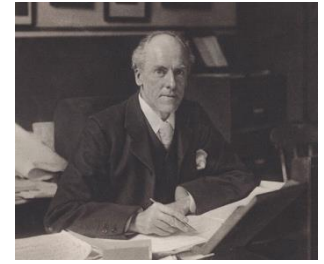
Karl Pearson



Ronald Fisher

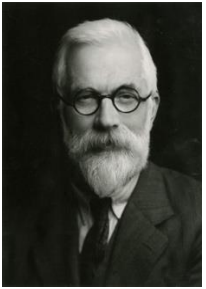
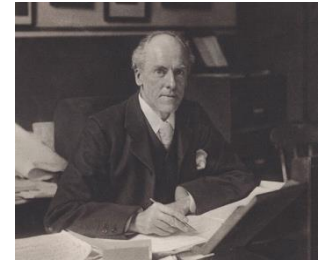
P Values

- “The probability of obtaining the data given the null hypothesis is true”

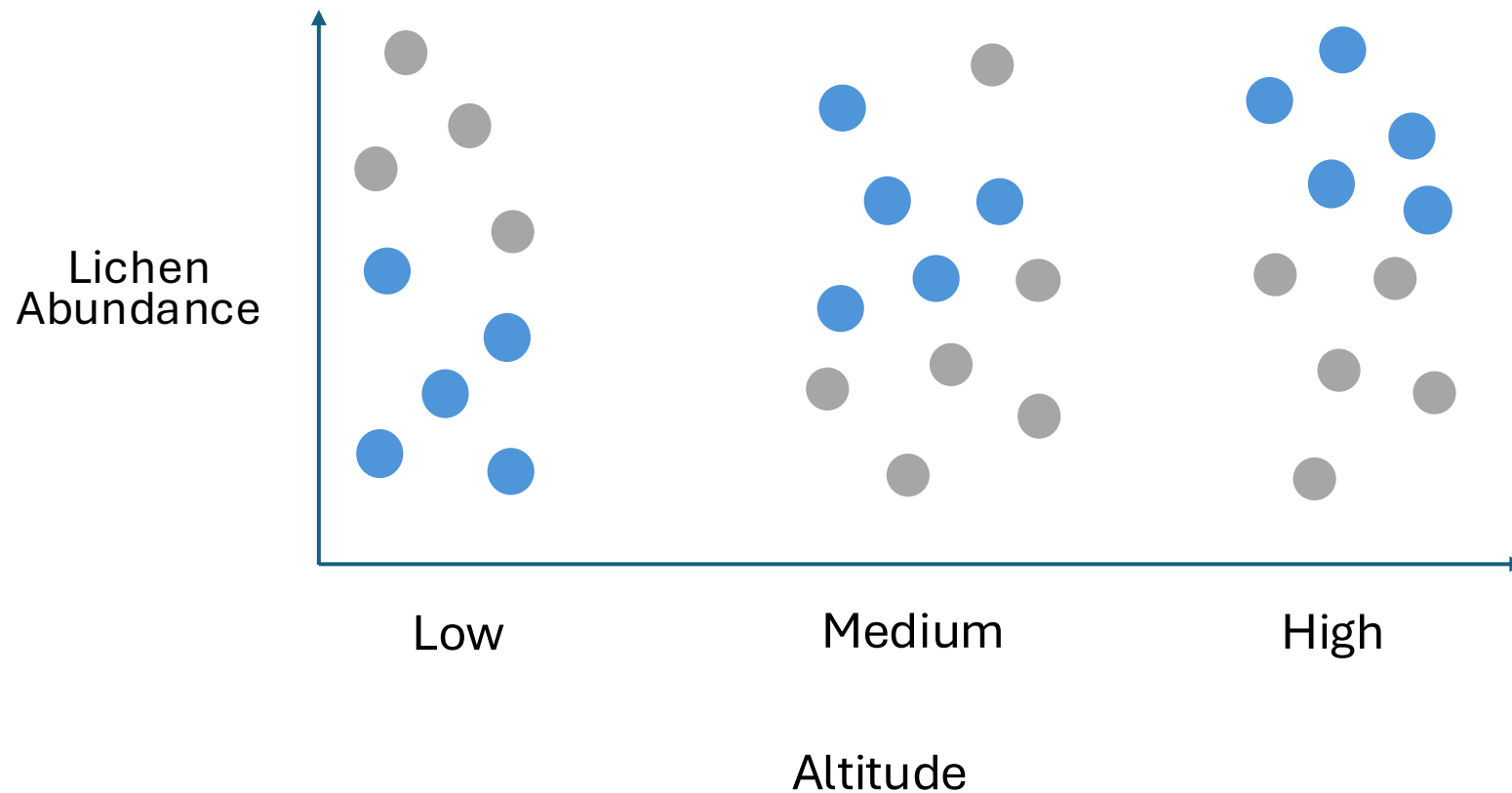


P Values

- “The probability of obtaining the data given the null hypothesis is true”



*Gentleman, this
isn't very intuitive*

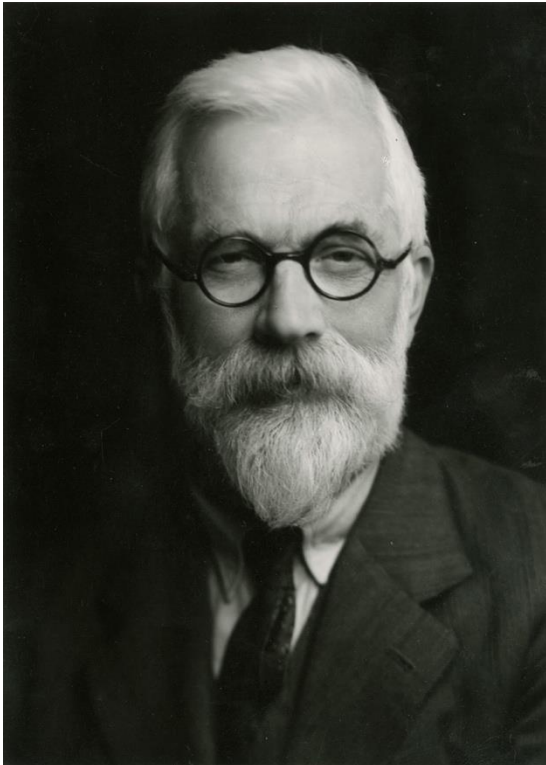


P Values

- Where did $P < 0.05$ come from?

P Values

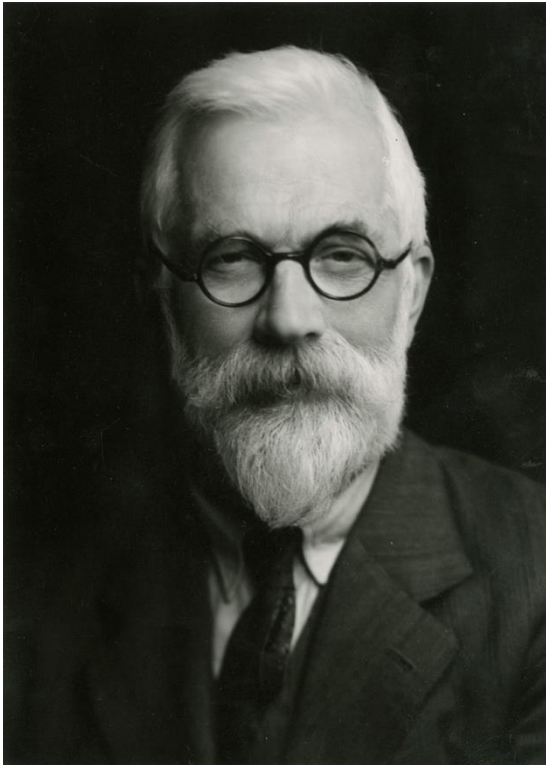
- Where did $P < 0.05$ come from?



“The value for which $P=0.05$, or 1 in 20... is convenient to take this point as a limit in judging whether a deviation ought to be considered significant or not”

P Values

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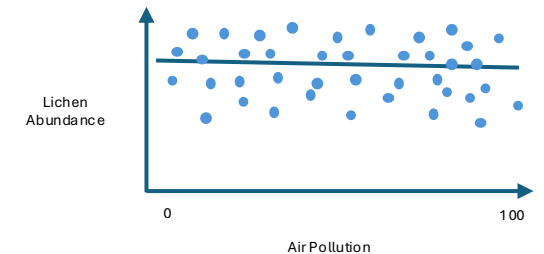


“The value for which $P=0.05$, or 1 in 20... is convenient to take this point as a limit in judging whether a deviation ought to be considered significant or not”

i.e. it's totally arbitrary

P Values

- **P Values can often tell you there's a significant relationship when there isn't in reality (esp if you do lots of tests)**
- **P values are not good at showing effect size (esp with large datasets)**
- **Please please read this excellent article:**
<https://www.nature.com/articles/506150a>



On other ways of doing Statistics

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ANOVA

Linear Model



Frequentist Approaches, based
on null hypothesis testing

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Frequentist Statistics

ANOVA

Linear Model

(in the way we teach it)

Linear Model

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Frequentist Statistics

ANOVA

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Many other things

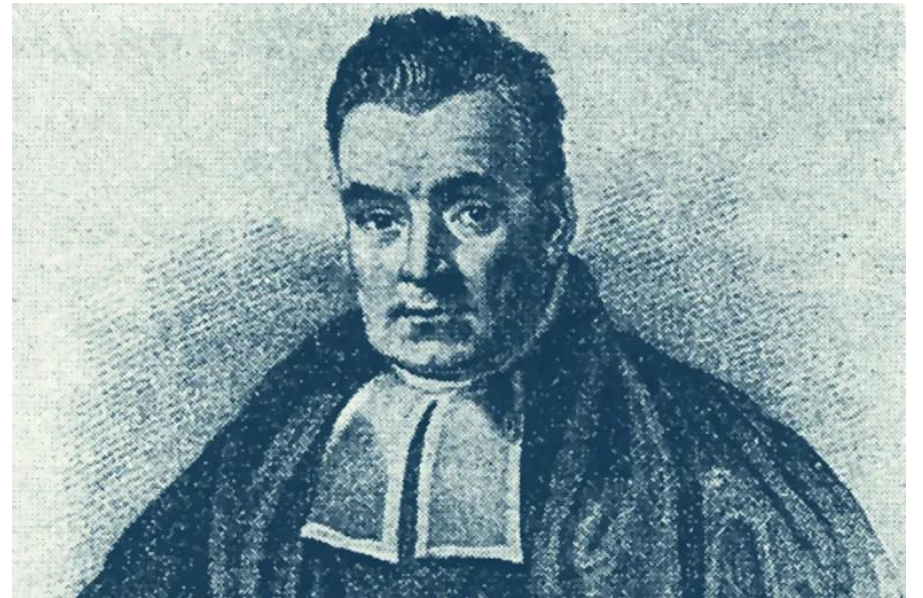
Bayesian Statistics

Bayesian Statistics

- Where Frequentist Stats asks what's “the probability of obtaining the data given the null hypothesis is true”
- Bayesian Stats just asks “what's the probability that this happened, in reality?” (i.e. much more intuitive!)
- Doesn't contain p-values

Bayesian Statistics

- **Started by Thomas Bayes, who wrote a letter proposing his theory**
- **... then he died**
- **In 1763, his friend published the letter**



Bayesian Statistics

- Theory picked up by Pierre—
Simon Laplace, who developed
it into the early 1800s.



Bayesian Statistics

- You can give your model 'prior' information, about e.g. how plausible your hypothesis is, or what you think values might be

Bayesian Statistics

- Which is better?
- Largely personal preference
- This clever man says it's a question of being an optimist vs a pessimist: <https://www.youtube.com/watch?v=HUAЕ26lNDuE>
 - Frequentist: What are the chances I'm wrong, are we likely to get a similar result if we collect this data again?
 - Bayesian: What are the chances I'm right, what can I learn from the data I have?

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Frequentist Statistics

ANOVA

Linear Model
(in the way we teach it)

Bayesian Statistics

Bayesian Models

(many types – often that do the ‘same thing’ as frequentist models, just in a different framework)

Information
Theory

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ANOVA

Linear Model
(in the way we teach it)

Bayesian Statistics

Bayesian Models

(many types – often that do the ‘same thing’ as frequentist models, just in a different framework)

Information Theory

“How much information does a variable give us?”
Includes AIC comparison – to be learnt about later!

To Conclude

- **Try not to be scared by stats. It's confusing because it's confusing, not because there's something wrong with you**
- **Any of us can figure out! (it just takes a bit of time)**

To Conclude

- **Try not to be scared by stats. It's confusing because it's confusing, not because there's something wrong with you**
- **Any of us can figure out! (it just takes a bit of time)**
- **Keep your wits about you – look at graphs of your data, think critically and logically. Don't be tempted to just blindly trust a test – they were invented by flawed grumpy men, and are themselves flawed!**
- **Does my result make sense? Is my result useful?**